

1 & 2.

<https://colab.research.google.com/drive/1lvGUgOzkH8ng0vN1jLlxeduAWyx6zxry?usp=sharing>

3. The data set is composed of 100 observations which represent the number of hours spent talking on cellphones by college students in one week. The mean time spent is 9.77 hours. The median is 10 hours, with a standard deviation of 1.97 hours which reflects moderate imbalance, with a range of 4 - 15 hours a week. The confidence interval for the population mean at 95% is (9.38, 10.16). The true mean of cell phone usage is most likely in this range.

When segregated by gender, male students have a mean of 9.82 hours with a standard deviation of 2.15 hours. On the other hand, female students have a mean of 9.72 hours with a standard deviation of 1.78 hours. The confidence interval for males is (9.21, 10.43), and for females, (9.21, 10.23).

Male phone usage averages 0.10 hours more a week. The confidence intervals suggest and conclude there is little evidence to support a large difference in average phone usage by male and female students in the sample.

11.) 1.) dataset: 2, 3, 7, 8, 10, 10, 12, 13, 17, 20
 no. of observation: $n = 10$

First: $\sum x = 2 + 3 + 7 + 8 + 10 + 10 + 12 + 13 + 17 + 20 = 104$

$$m'_1 = \frac{104}{10} = \boxed{10.4}$$

Second: x^2

X	x^2
2	4
3	9
7	49
8	64
10	100
10	100
12	144
13	169
17	289
20	400

$$\Rightarrow \sum x^2 = 1384$$

$$m'_2 = \frac{1384}{10} = \boxed{138.4}$$

Third: $x^3 \Rightarrow \sum x^3 = 21200 \Rightarrow m'_3 = \frac{21200}{10} = \boxed{2120}$

Fourth: $x^4 \Rightarrow \sum x^4 = 356516 \Rightarrow m'_4 = \frac{356516}{10} = \boxed{35651.6}$

2.) mean: $\bar{x} = 10.4$

First: $m_1 = 0$

Second: $m_2 = 30.64$

Third: $m_3 = 30.432$

Fourth: $m_4 = 1797.8272$

Formula: $m_r = \frac{1}{n} \sum (x - \bar{x})^r$

3.) Formula: $m_4 = m'_4 - 4m'_1 m'_3 + 6(m'_1)^2 m'_2 - 3(m'_1)^4$

$$35651.6 - 88192 + 89837.4144 - 351.1872$$

$$m_4 = 1797.8272 \rightarrow \text{Verified.}$$

$$11.1. > m'_4 = m_4 + 4hm_3 + 6h^2m_2 + h^4$$

$$\text{Where: } h = m'_1$$

$$\Rightarrow x = (x - \bar{x}) + \bar{x}$$

$$\Rightarrow x^4 = (x - \bar{x})^4 + 4\bar{x}(x - \bar{x})^3 + 6\bar{x}^2(x - \bar{x})^2 + 4\bar{x}^3(x - \bar{x}) + \bar{x}^4$$

$$\Rightarrow m'_4 = m_4 + 4\bar{x}m_3 + 6\bar{x}^2m_2 + 4\bar{x}^3m_1 + \bar{x}^4$$

$$m_1 = 0 \text{ so, } m'_4 = m_4 + 4\bar{x}m_3 + 6\bar{x}^2m_2 + \bar{x}^4$$

$$\text{since, } h = m'_1 = \bar{x}$$

$$\text{so, } m'_4 = m_4 + 4hm_3 + 6h^2m_2 + h^4 \rightarrow \text{Proved}$$

$$1V. -t_1, t_1 = 0.90, v = 20$$

$$\text{two tails} = 1 - 0.90 = 0.10$$

$$\text{Each} = 0.05$$

$$\rightarrow t_1 = t_{0.05, 20}$$

$$t_1 = 1.725$$

$$2. > t_{0.025, 20}$$

$$\rightarrow t_1 = 2.086$$

thus

$$\rightarrow -t_1 = -2.086$$

$$3. > 0.005$$

$$\rightarrow t_1 = t_{0.005, 5}$$

$$t_1 = 4.032$$

$$4. > \text{Area right of } t_1 = 0.55, v = 16$$

$$\text{left} = 1 - 0.55 = 0.45$$

$$t_1 = t_{0.45, 16}$$

$$t_1 \approx -0.13$$